

FinGPT Assignment 2: Effect of Overfitting Control in Fine-Tuning Llama 3.1 and DeepSeek Models

Tiantian Hang

th3166

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Abstract

This report investigates how training-parameter adjustments targeting overfitting influence the fine-tuning outcomes of two large language models (LLMs) within the FinGPT framework: **Llama 3.1 (8B)** and **DeepSeek R1-Distill-Llama (8B)**. Both models were fine-tuned on the Dow Jones 30 dataset using the Low-Rank Adaptation (LoRA) approach. Two experimental rounds were conducted: Round 1 served as a baseline fine-tuning configuration, and Round 2 introduced regularization techniques—lower learning rates, larger warm-up ratios, cosine scheduling, and reduced LoRA capacity—to mitigate overfitting. Results demonstrate that these modifications improved stability and generalization for Llama 3.1 but slightly reduced DeepSeek’s fine-tuning accuracy, indicating that the effect of overfitting control is model-dependent.

1 Introduction

Large Language Models (LLMs) have become indispensable tools in financial text understanding and reasoning. However, fine-tuning them on compact, homogeneous datasets can lead to severe overfitting, harming out-of-domain generalization. This study examines two FinGPT fine-tuned variants—FinGPT-Llama3 and FinGPT-DeepSeek—each trained twice under different hyperparameter regimes. By contrasting the two fine-tuning rounds for each model, we aim to reveal how deliberate regularization shapes predictive robustness, textual coherence, and inference efficiency.

2 Dataset and Task

The dataset `FinGPT/fingpt-forecaster-dow30-202305-202405` consists of recent news and fundamental indicators for the Dow Jones 30 companies. Each input prompt requests a structured analysis containing three sections: (1) positive developments, (2) potential concerns, and (3) next-week price prediction with rationale. Ground-truth responses are generated by GPT-4. Evaluation combines textual and numerical dimensions through ROUGE, binary direction accuracy, and mean-squared error (MSE).

3 Fine-Tuning Setup

Both models were fine-tuned using PEFT-LoRA in `bfloat16` precision on an Nvidia A100GPU. Adapters targeted attention and feed-forward layers, with sequence length limited to 2048. Table 1 summarizes the key configurations of the two fine-tuning rounds.

Table 1: Fine-Tuning Configurations for the Two Experimental Rounds

Parameter	Llama R1	Llama R2	DeepSeek R1	DeepSeek R2
Learning Rate (η)	5e-5	2e-5	5e-5	3e-5
Epochs (T)	3	5	3	4
Warm-Up Ratio	0.03	0.10	0.03	0.05
Scheduler	constant	cosine	constant	cosine
LoRA Rank (r)	default	8	default	6
LoRA α	default	16	default	12
Batch Size	1	1	1	1
Grad. Accum. Steps	16	16	16	16
Max Seq. Len.	2048	2048	2048	2048

4 Regularization Strategy

Round 2 incorporated the following anti-overfitting adjustments:

1. Lower learning rate and cosine decay to produce smoother parameter updates.
2. Higher warm-up ratio to stabilize early optimization.
3. Reduced LoRA rank and α to constrain adapter capacity.
4. Identical prompt templates and truncation rules to ensure consistent data distribution.

5 Evaluation Metrics

Each model’s Round 1 and Round 2 versions were tested on the same held-out Dow30 set (300 samples). Metrics include:

- **Binary Accuracy (Acc)** – Correct prediction of price-direction sign.
- **Mean Squared Error (MSE)** – Deviation between predicted and actual percentage changes.
- **ROUGE Scores (1/2/L)** – Textual similarity for “Positive Developments,” “Potential Concerns,” and “Analysis.”
- **Inference Latency** – Average generation time per sample.

6 Results: Impact of Overfitting Control

6.1 Overall Comparison

Table 2 summarizes quantitative changes from Round 1 to Round 2 for both fine-tuned models.

Table 2: Performance Comparison of Fine-Tuned Models Between Round 1 and Round 2

Model	Round	Binary Acc.	MSE	Avg Latency (s)
FinGPT-Llama3	R1	0.429	11.83	24.98
FinGPT-Llama3	R2	0.495	11.94	25.01
FinGPT-DeepSeek	R1	0.471	11.74	24.23
FinGPT-DeepSeek	R2	0.520	8.79	24.11

6.2 FinGPT-Llama3

After regularization, FinGPT-Llama3’s binary accuracy increased from 0.429 to 0.495, reflecting improved directional prediction stability. The MSE remained essentially unchanged ($11.83 \rightarrow 11.94$), indicating that the model’s magnitude estimation was unaffected. Inference latency stayed constant (25 s/sample). Overall, the overfitting-controlled setup enhanced generalization without sacrificing efficiency.

6.3 FinGPT-DeepSeek

For FinGPT-DeepSeek, the binary accuracy rose modestly ($0.471 \rightarrow 0.520$), but the MSE decreased substantially ($11.74 \rightarrow 8.79$), suggesting better calibration of predicted price

changes. Despite a smaller LoRA capacity, the model maintained inference speed and textual fluency, with ROUGE scores around 0.43. These results indicate that the cosine-scheduled, lower-learning-rate configuration helped stabilize DeepSeek’s fine-tuning performance.

6.4 Cross-Model Observation

Comparing the two LLM families, both benefited from overfitting mitigation but in different ways: Llama 3.1 gained directional consistency, while DeepSeek improved numerical accuracy. Neither model exhibited any increase in inference time, confirming that enhanced regularization incurs negligible computational overhead.

7 Discussion and Conclusion

This two-round fine-tuning experiment demonstrates the tangible effects of regularization in financial LLM adaptation. Applying smaller learning rates, larger warm-ups, and cosine scheduling improved generalization stability for both FinGPT-Llama3 and FinGPT-DeepSeek. While the absolute magnitude of improvement varied, both models achieved better balance between prediction accuracy and textual quality without added latency.

Future work may integrate:

- Weight decay, dropout, and early stopping for stronger regularization;
- Multi-stage fine-tuning separating textual and numerical objectives;
- Reinforcement-based optimization directly targeting binary accuracy and MSE.